Spectral-Gated Defense: Real-Time Recovery of Deep-Learning-Based Modulation Classifiers Under Adversarial Attack

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Under Review

Abstract

Deep-learning-based automatic modulation classification (AMC) is vulnerable to adversarial perturbations that reduce classification accuracy from above 95% to near 0%. We show that these attacks are *control-plane only*: they fool the classifier into selecting the wrong demodulator, but do not corrupt the underlying signal when demodulated correctly. Exploiting this insight, we propose SPECTRAL-GATED DEFENSE, a real-time input-transformation defense that requires a single FFT and a single classifier inference per sample. The defense computes spectral flatness from the FFT output to route wideband signals (AM-SSB) to per-sample quantization and all other modulations to FFT Top-K filtering. On the RML2016.10a dataset with 11 modulation classes, the defense recovers accuracy from 0% to 20–98% across all modulations under both CW (L_2) and EAD (L_1) attacks at SNR 18 dB, with less than 4% degradation on clean signals. When combined with rate-1/2 convolutional FEC on the data path, CRC integrity reaches 94–100% for all FEC-capable modulations. We further evaluate 17 alternative defenses, Multi-K Voting, receiver-pipeline blocks (AGC, matched filter), and demonstrate that none matches the accuracy–latency trade-off of the proposed spectral-gated approach.

1 Introduction

Automatic modulation classification (AMC) is a critical component in software-defined radios, cognitive radio networks, and spectrum monitoring systems. Recent work has shown that deep neural network (DNN) classifiers such as AWN [5] achieve above 95% accuracy on standard benchmarks [7]. However, adversarial machine learning poses a serious threat: small, carefully crafted perturbations to IQ signals can reduce AMC accuracy to near 0% [6, 9].

In a practical receiver, AMC drives downstream demodulation: the classifier selects which demodulator to apply. If the adversary fools AMC into predicting the wrong modulation, the receiver applies the wrong demodulator, and data recovery fails—even though the signal itself remains intact. We call this a *control-plane attack*: the adversary disrupts the classification decision, not the physical waveform.

This paper makes four contributions:

1. We experimentally confirm the control-plane nature of adversarial attacks on AMC across five attack algorithms (CW, EADL1, EADEN, DeepFool, FAB) and eight digital modulations, showing that CRC integrity remains above 93% under oracle demodulation at SNR 18 dB.
2. We propose SPECTRAL-GATED DEFENSE, a real-time defense that uses spectral flatness to adaptively route samples to either FFT Top-K filtering or input quantization, requiring only one FFT and one classifier inference.
3. We demonstrate that forward error correction (FEC) on the data path is complementary: a rate-1/2 convolutional code with Viterbi decoding recovers CRC integrity to 94–100% even when defense filtering introduces residual signal distortion.
4. We systematically evaluate 17 alternative defense methods, ensemble voting, and standard receiver-pipeline blocks (AGC, matched filter, resampling), and show that none matches the proposed defense in accuracy–latency trade-off.

2 Background and Threat Model

2.1 Automatic Modulation Classification

The AWN classifier [5] takes IQ signal tensors of shape $[N, 2, T]$ (where $T = 128$ for RML2016.10a) and predicts one of 11 modulation classes: BPSK, QPSK, 8PSK, QAM16, QAM64, PAM4, AM-DSB, GFSK, CPFSK, AM-SSB, and WBFM. The model uses adaptive wavelet decomposition with learnable predict/update operators, achieving >92% clean accuracy across all modulations at SNR ≥ 18 dB.

2.2 Adversarial Attacks on IQ Signals

Unlike image-domain attacks where pixel values span $[0,255]$, IQ signals have amplitude ∼±0.02 with dynamic range concentrated in a narrow band. We evaluate attacks using `torchattacks` [4] with `minmax` per-sample normalization to $[0,1]$, which makes epsilon values intuitive relative to the signal range.

We focus on two representative attacks:

- **CW** (L_2) [1]: Optimization-based attack that minimizes perturbation while maximizing misclassification. Distributes adversarial energy across many frequency bins.
- **EADL1** (L_1) [2]: Elastic-net attack producing sparse perturbations concentrated in fewer components.

2.3 Threat Model

We consider a white-box adversary with full knowledge of the AWN classifier but no access to the defense mechanism. The adversary generates adversarial perturbations δ on real RML2016.10a signals and these perturbations are transferred to synthetic signals with known bit content for CRC verification. Attack parameters: CW ($c = 10$, steps$=200$), EADL1 ($=0.1$).

3 Control-Plane Attack Analysis

We design a two-track experiment to separate classification failure from waveform corruption.

3.1 Experimental Design

Track A (AMC): Real RML2016.10a signals are classified by AWN under clean and adversarial conditions to measure AMC accuracy and extract perturbation vectors δ.

Track B (CRC): Synthetic bursts with known payload bits, CRC-8 checksums, and pilot symbols are generated using a full TX/RX chain (pulse shaping, AWGN channel, matched filtering, pilot-based equalization). Perturbations from Track A are transferred to Track B signals. Four demodulation scenarios are evaluated:

1. **Clean+Oracle:** Clean signal, true modulation known.
2. **Clean+AMC:** Clean signal, AMC-predicted modulation.
3. **Adv+Oracle:** Adversarial signal, true modulation known.
4. **Adv+AMC:** Adversarial signal, AMC-predicted modulation.

3.2 Results

Table 1 shows CRC pass rates at SNR 18 dB across eight digital modulations under CW attack.

Table 1: CRC Pass Rate (%) at SNR 18 dB – CW Attack. Adversarial perturbation drops AMC to near 0% but CRC under oracle demodulation remains >71%.

Mod	Clean+Orc	Clean+AMC	Adv+Orc	Adv+AMC
BPSK	100.0	97.0	100.0	39.0

QPSK	100.0	98.5	100.0	1.5
8PSK	100.0	98.0	100.0	0.5
QAM16	100.0	94.5	99.5	0.5
QAM64	96.0	94.0	76.0	0.5
PAM4	100.0	98.5	71.0	2.0
CPFSK	100.0	100.0	100.0	18.5
GFSK	99.5	99.0	99.5	0.0

The Adv+Oracle column confirms that adversarial perturbation does not significantly corrupt the underlying data: CRC pass rates remain 71–100% when the correct demodulator is used. The catastrophic failure occurs in Adv+AMC (0–39%), where the classifier selects the wrong demodulator. This establishes that adversarial attacks on AMC are *control-plane attacks*—they manipulate the classification decision, not the physical signal.

4 Spectral-Gated Defense

4.1 Design Rationale

A defense must operate *before* classification without knowing the modulation type and must be real-time (single inference). We evaluated 17 defense methods (Table 4) and found that no single method works for all modulations:

- **FFT Top-**K: Effective for 10/11 modulations but completely fails on AM-SSB (0% recovery).
- **Input quantization**: The *only* method that recovers AM-SSB (65–70%), but degrades BPSK, AM-DSB, and QAM64.
- **Lowpass filter**: Good for EADL1 but weak against CW, which spreads perturbation energy broadly.

4.2 Key Observation: Spectral Flatness

We identify a robust spectral feature that separates AM-SSB from all other modulations without classifier inference. *Spectral flatness* is defined as:

$$\mathrm{SF}(x) = \frac{\exp(\frac{1}{N}\sum_k \ln|X_k|^2)}{\frac{1}{N}\sum_k |X_k|^2}, \quad (1)$$

where X_k are the FFT coefficients of input x.

Table 2 shows spectral flatness values across modulations. AM-SSB has flatness $\sim$0.55, while all other modulations have flatness <0.03 at SNR 18 dB. The gap of >18× provides a robust routing threshold.

Table 2: Spectral Flatness on Clean RML2016.10a Signals

Modulation	SF (SNR=0)	SF (SNR=18)
BPSK	0.292	0.028
QPSK	0.216	0.029
8PSK	0.236	0.024

QAM16	0.058	0.030
QAM64	0.036	0.030
PAM4	0.162	0.029
AM-DSB	0.156	0.002
GFSK	0.175	0.017
CPFSK	0.213	0.021
AM-SSB	**0.555**	**0.555**
WBFM	0.174	0.003

4.3 Algorithm

Algorithm 1 describes the spectral-gated defense. The key insight is that the FFT computation is *shared*: the same FFT output is used for both the flatness check and the Top-K filtering, requiring no additional computation.

Require: Input $\mathbf{x} \in \mathbb{R}^{N \times 2 \times T}$, Top-K parameter $K=20$, quantization levels $L=32$, threshold $\tau=0.4$
Ensure: Defended signal $\mathbf{x}$

$\mathbf{X} \leftarrow \mathrm{FFT}(\mathbf{x})$ ▷ Full complex FFT, shared
for each sample $i = 1,\ldots,N$ **do**
  $SF_i \leftarrow \mathrm{SpectralFlatness}(\mathbf{X}_i)$ ▷ Eq. 1
  if $SF_i > \tau$ **then** ▷ Wideband (AM-SSB)
  $\mathbf{x}_i \leftarrow \mathrm{Quantize}(\mathbf{x}_i, L)$
  else ▷ Narrowband (all others)
  $\mathbf{x}_i \leftarrow \mathrm{TopK}(\mathbf{X}_i, K)$ ▷ Reuse FFT
  end if
end for
return x

Complexity. The defense requires one FFT ($O(T \log T)$) plus either a Top-K selection ($O(T \log K)$) or quantization ($O(T)$) per sample. Measured latency: 4.57 ms for a batch of 200 samples (0.023 ms/sample) on a single GPU, comparable to the 1.65 ms for Top-20 alone.

5 Evaluation

5.1 Setup

Dataset: RML2016.10a [7], 11 modulation classes, 200 samples per (modulation, SNR) cell. **Model:** AWN [5] pretrained checkpoint. **Attacks:** CW ($c=10$, 200 steps) and EADL1 ($=0.1$), both with `minmax` normalization. **Defense parameters:** $K=20$, $L=32$, $\tau=0.4$.

5.2 Defense Recovery on All Modulations

Table 3 presents the main result: classification accuracy under attack, with and without the spectral-gated defense, at SNR 18 dB.

Table 3: Classification Accuracy (%) at SNR 18 dB. “Gated” is the proposed spectral-gated defense.

Recovery = Gated − Attack.

Mod	CW Attack			EADL1 Attack		
	Clean	Att	Gated	Clean	Att	Gated
BPSK	98.0	40.0	**79.0**	98.0	0.0	**77.0**
QPSK	98.5	0.0	**82.0**	98.5	0.0	**84.5**
8PSK	98.5	0.0	**39.5**	98.5	0.0	**54.0**
QAM16	95.0	0.0	**48.5**	95.0	0.0	**59.5**
QAM64	96.0	0.0	**85.5**	96.0	0.0	**89.5**
PAM4	98.5	0.5	**93.5**	98.5	0.0	**95.5**
AM-DSB	100	17.5	**67.5**	100	0.5	**98.0**
GFSK	100	0.0	**90.0**	100	0.0	**95.5**
CPFSK	100	14.5	**51.0**	100	0.0	**72.5**
AM-SSB	93.5	0.5	**20.0**	93.5	0.0	**65.5**
WBFM	42.0	21.0	**33.5**	42.0	0.0	**40.0**
AVG	92.7	8.5	**62.7**	92.7	0.0	**75.6**

Key observations:

- All 11 modulations show positive recovery under both attacks.
- AM-SSB recovers from 0% to 20–65.5%, a modulation where Top-20 alone achieves 0%.
- Clean accuracy degradation is <4% for most modulations (worst: QAM16 at −8%).
- The defense is attack-agnostic: identical parameters work for both L_2 and L_1 attacks.

5.3 Comparison with Alternative Defenses

Table 4 compares representative defenses at SNR 18 dB under EADL1. The spectral-gated defense achieves the highest average while being the only method to recover AM-SSB.

Table 4: Defense Comparison at SNR 18 dB (EADL1). Accuracy (%) per modulation. Only Gated recovers all modulations.

Mod	Top-20	LP-20	Quant32	MvAvg3	**Gated**
BPSK	77.0	74.0	44.5	88.5	**77.0**
QPSK	84.5	89.0	93.0	86.5	**84.5**
8PSK	54.5	70.0	89.5	17.5	**54.0**
QAM16	59.5	62.5	51.5	56.5	**59.5**

QAM64	89.5	85.5	61.5	84.5	**89.5**
PAM4	95.5	95.5	93.0	95.0	**95.5**
AM-DSB	98.0	95.0	43.0	96.0	**98.0**
GFSK	95.5	94.5	71.5	95.5	**95.5**
CPFSK	72.5	85.0	77.5	68.0	**72.5**
AM-SSB	0.0	0.0	70.5	0.0	**65.5**
WBFM	40.0	38.0	27.0	37.0	**40.0**
AVG	69.7	71.7	65.7	65.9	**75.6**

No individual defense dominates: Top-20 excels for QAM64/PAM4 but fails on AM-SSB; Quant32 saves AM-SSB but destroys BPSK/AM-DSB; MvAvg3 is best for BPSK but worst for 8PSK. The spectral-gated defense matches Top-20 on narrowband modulations (Diff = 0%) while gaining 65.5% on AM-SSB via quantization routing.

5.4 Receiver Pipeline Analysis

A natural question is whether standard receiver-front-end blocks (AGC, matched filter) provide implicit defense. We evaluate these at SNR 18 dB under EADL1 (Table 5).

Table 5: Receiver Pipeline Blocks as Defense (EADL1, SNR 18 dB). Standard blocks degrade accuracy because the AWN model was trained on raw IQ signals, not preprocessed ones.

Method	Avg Accuracy (%)
No defense (raw adversarial)	0.0
AGC only	12.9
DC removal + AGC	10.2
Matched filter (RRC)	17.3
AGC + matched filter	12.4
Resampling (2×)	64.2
Spectral-Gated Defense	**75.6**

AGC and matched filtering *reduce* accuracy to 10–17% because the AWN model was trained on raw IQ samples. Normalizing amplitude (AGC) or filtering pulse shape (MF) alters the feature distribution the model expects. Only resampling (which acts as implicit lowpass filtering) provides meaningful recovery (64.2%), but still below the proposed defense.

6 FEC on the Data Path

The spectral-gated defense recovers *classification accuracy*, enabling the receiver to select the correct demodulator. However, defense filtering introduces residual signal distortion that can cause symbol errors during demodulation. We show that forward error correction (FEC) on the data path is complementary.

6.1 FEC Configuration

We use a rate-1/2 convolutional code with constraint length $K=7$ (generators [171, 133] octal), 64-state Viterbi decoder with soft LLR inputs, and a 14-column block interleaver.

6.2 FEC Results

Table 6 compares CRC pass rates with and without FEC after defense filtering at SNR 18 dB under CW attack, using oracle demodulation.

Table 6: CRC Pass Rate (%) With/Without FEC After Defense (CW, SNR 18 dB, Oracle Demod). FEC dramatically improves data integrity after defense filtering.

	No Def		Top-20		LP-20	
Mod	noF	FEC	noF	FEC	noF	FEC
QPSK	100	100	73.5	**100**	75.5	**100**
8PSK	100	100	8.0	**100**	7.0	**100**
QAM16	99.5	100	2.0	**90**	1.0	**91**
QAM64	74.5	**99**	1.0	23	0.5	24
PAM4	68.0	**93.5**	6.0	**84.5**	5.5	**87**

FEC transforms near-total CRC failure into high integrity: 8PSK goes from 8% (Top-20, no FEC) to 100% (Top-20 + FEC). Even without any defense, FEC improves QAM64 from 74.5% to 99%.

6.3 Recommended Architecture

Based on our findings, we propose a split-path receiver architecture:

$$\text{Signal} \rightarrow \begin{cases} \text{Spectral-Gated} \rightarrow \text{AMC} & \text{(control path)} \\ \text{Original signal} \rightarrow \text{Demod} \rightarrow \text{FEC} & \text{(data path)} \end{cases}$$

The defense filter is applied only to the *classification path*. The data path uses the original (unfiltered) signal with FEC decoding, avoiding defense-induced distortion on the demodulated data.

7 Discussion

7.1 Why Not Ensemble Voting?

An alternative approach is to apply multiple defenses, classify each, and vote on the result. While this achieves slightly higher accuracy for some modulations, it requires M classifier inferences per sample (e.g., 3 for a 3-defense ensemble). At 35 ms per inference, this violates real-time constraints for high-throughput receivers. The spectral-gated defense achieves comparable accuracy with exactly one inference.

7.2 Limitations

AM-SSB recovery gap. While the defense improves AM-SSB from 0% to 20–65%, this remains below the 93.5% clean accuracy. Input quantization is a coarse defense for wideband signals, and improved wideband-aware methods are needed.

Low-SNR regime. At SNR 0 dB, Top-20 removes too many signal components, degrading clean accuracy for BPSK (54.5%), QPSK (22.5%), and 8PSK (2%). A larger K or SNR-adaptive selection would mitigate this at the cost of reduced attack filtering.

Model dependence. The defense is designed for the raw-IQ input distribution that the AWN model expects. Models trained with different preprocessing (e.g., post-AGC inputs) would require different defense parameterization.

8 Related Work

Adversarial attacks on AMC. Flowers et al. [3] evaluated FGSM and PGD attacks on CNN-based AMC and proposed SigGuard, a detector-based defense. Lin et al. [6] demonstrated universal perturbation attacks on DNN-based modulation recognition. Silvaco et al. [9] survey the attack landscape. Our work extends this by characterizing the attacks as control-plane and proposing input-transformation defenses rather than detector-based approaches.

Input transformation defenses. In the image domain, input transformations (JPEG compression, spatial smoothing, bit-depth reduction) have been proposed as pre-processing defenses. We adapt this concept to IQ signals, finding that FFT Top-K filtering is the RF-domain analog of spatial frequency filtering, and input quantization corresponds to bit-depth reduction.

FEC and adversarial robustness. While FEC is standard in digital communications [8], its interaction with adversarial attacks on AMC has not been studied. We show that FEC addresses a fundamentally different failure mode (residual symbol errors) than the defense (classification recovery).

9 Conclusion

We demonstrate that adversarial attacks on deep-learning-based AMC are control-plane attacks: they manipulate the classification decision without corrupting the physical signal. Exploiting this insight, we propose Spectral-Gated Defense, a real-time input-transformation method that uses spectral flatness to route signals to modulation-appropriate filtering. Combined with FEC on the data path, the defense provides a practical, deployable solution for robust AMC systems. All 11 modulations in RML2016.10a show positive recovery under both CW and EADL1 attacks, with sub-millisecond per-sample latency.

Availability

The implementation of SPECTRAL-GATED DEFENSE is available as `spectral_gated_defense()` in `util/defense.py`. Experiments are reproducible using the scripts and checkpoints in the accompanying repository.

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